

Aaron Zhang

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Robotics engineering student at the University of Michigan combining embedded systems and mechanical design experience with a design-driven, cross-disciplinary background in visual art.

EDUCATION

University of Michigan

Bachelor of Science in Engineering in Robotics

Coursework: Robot Operating Systems, Circuit Analysis, Data Structures & Algorithms (C++), Robot Localization

Ann Arbor, MI

May 2027

SKILLS

Languages: Python, C++, MATLAB, JavaScript, Java, HTML/CSS, TypeScript

Frameworks/Tools: PID Control, SLAM, React, Node.js, PyTorch, REST APIs, Tkinter | Git, Excel, Notion, Obsidian, Cursor

Certifications: Google Agile Essentials, Google AI Essentials, Google Foundations of UX Design

WORK EXPERIENCE

Themelios Education – STEM Instructor and Youth Mentor

Chicago, IL *June 2025 – August 2025*

- Designed and led 10+ hands-on robotics/STEM workshops using Lego SpikePrime Robotics for 50+ students in grades 8-12, teaching sensor I/O, motor control, programming logic to help build early interest in engineering pathways
- Programmed and debugged Python/MATLAB test cases for robotics projects covering sensor input processing and motor-control logic, achieving a ~40% improvement in learner performance

Rutech Robotics Inc. – Robotics Technician

Chicago, IL *May 2024 – August 2024*

- Diagnosed and repaired hardware/software faults across 25+ service robots, using RMS teleoperation and Android ADB to reduce repeat service calls by 25% and customer malfunction tickets by 35%
- Configured, mapped, and troubleshooted service robots across 20+ client deployments, communicating resolutions to technical and non-technical stakeholders to support successful deployment and robot operation

AI Data Annotation Specialist | TELUS | Appen

Remote *April 2024 – Current*

- Annotated and validated 2250+ data entries across 25+ large-scale datasets for ML models maintaining 97% accuracy

PROJECT EXPERIENCE

M314 Aliens Motion Tracker | Independent Project | Arduino/ESP32, C++, Embedded Systems

July 2026 – September 2026

- Designed and built a functional prop replica of the *Aliens* M314 motion tracker, integrating motion sensing with real-time audio-visual feedback (proximity beeping + LED/display readout)
- Programmed embedded control logic in C++ to process sensor input and drive proportional audio pulse-rate and visual output based on detected distance, achieving 10m detection range at 200ms response latency
- Engineered a custom enclosure balancing functional component layout with faithful prop design, applying a fabrication/visual-design background to the build

Lego Functioning Desert Eagle | Independent Project | Mechanical Design, Kinematics, Lego Technic

June 2026 – July 2026

- Designed and built a functioning scale replica of a Desert Eagle pistol in Lego Technic, engineering a working magazine release, spring-loaded firing mechanism, slide lock, and working slide-action mechanism using elastic actuation
- Iterated magazine design, elastic-tensioning configuration across 5+ prototypes to achieve reliable slide cycling
- Applied principles of leverage, elastic potential energy, spring tension, linkage kinematics to replicate real firearm mechanics within Technic parts constraints

PaperWeight | Obsidian Plugin | Independent Project | TypeScript, CodeMirror 6

July 2026 – September 2026

- Built and published an *Obsidian* plugin implementing a state-synchronization system that preserves UI state across real-time document mutations, addressing a personal workflow gap with no public API or existing plugin solution available
- Reverse-engineered undocumented internal APIs across competing subsystems to isolate state-to-render pipeline and resolve coordinate-frame drift and event-ordering bugs, enabling reliable UI-state preservation across a 180+ element system

Astralite: Minimalist Habit Tracking System | Independent Project | React, Node.js, UI/UX Design

March 2026 – June 2026

- Developed a full-stack habit-tracking web app for minimalists to address low consistency on personal goal-tracking, implementing gamified mechanics, and responsive progress visualization to reinforce consistent user behavior
- Designed a highly customizable UI and iterated UI/UX flows across 75+ feature revisions, improving UI readability by 35%